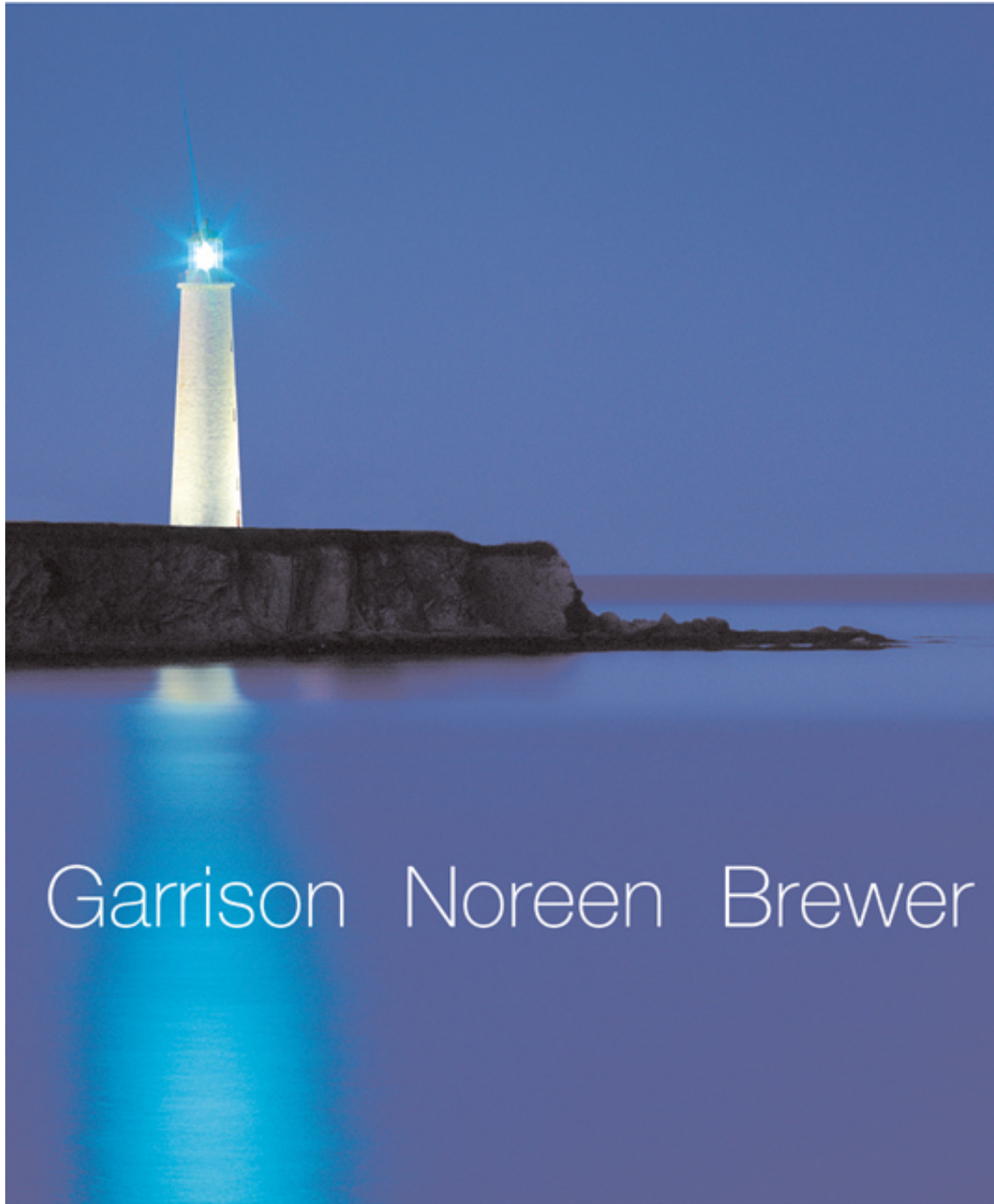




Garrison Noreen Brewer

11th Edition Chapter 6

Cost-Volume-Profit Relationships

Chapter Six

Basics of Cost-Volume-Profit Analysis



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1	<table><tr><th colspan="2">Racing Bicycle Company Contribution Income Statement For the Month of June</th></tr><tr><td>Sales (500 bicycles)</td><td>\$ 250,000</td></tr><tr><td>Less: Variables expenses</td><td>150,000</td></tr><tr><td>Contribution margin</td><td>100,000</td></tr><tr><td>Less: Fixed expenses</td><td>80,000</td></tr><tr><td>Net income</td><td>\$ 20,000</td></tr></table>						Racing Bicycle Company Contribution Income Statement For the Month of June		Sales (500 bicycles)	\$ 250,000	Less: Variables expenses	150,000	Contribution margin	100,000	Less: Fixed expenses	80,000	Net income	\$ 20,000
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Contribution Margin (CM) is the amount remaining from sales revenue after variable expenses have been deducted.

Basics of Cost-Volume-Profit Analysis



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CM is used first to cover fixed expenses. Any remaining CM contributes to net operating income.

The Contribution Approach

Sales, variable expenses, and contribution margin can also be expressed on a per unit basis. If Racing sells an additional bicycle, **\$200** additional CM will be generated to cover fixed expenses and profit.



The screenshot shows an Excel spreadsheet with a menu bar (File, Edit, View, Insert, Format, Tools, Data, Window, Help, Adobe PDF) and a toolbar. The active cell is M28. The spreadsheet contains a table titled "Racing Bicycle Company Contribution Income Statement For the Month of June". The table has three columns: Description, Total, and Per Unit. The data is as follows:

	Total	Per Unit
Sales (500 bicycles)	\$250,000	\$ 500
Less: Variables expenses	150,000	300
Contribution margin	100,000	<u>200</u>
Less: Fixed expenses	80,000	
Net income	<u>\$ 20,000</u>	

A red circle highlights the value "200" in the "Per Unit" column for "Contribution margin". A red arrow points from this circle to the text "\$200" in the paragraph above the spreadsheet.

The Contribution Approach

Each month Racing must generate at least **\$80,000** in total CM to break even.



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Racing Bicycle Company
Contribution Income Statement
For the Month of June

	Total	Per Unit
Sales (500 bicycles)	\$ 250,000	\$ 500
Less: Variables expenses	150,000	300
Contribution margin	100,000	\$ 200
Less: Fixed expenses	80,000	
Net income	\$ 20,000	

The Contribution Approach

If Racing sells 400 units in a month, it will be operating at the break-even point.



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	A	B	C	D	E	F	G	H																					
1	<table><tr><th colspan="3">Racing Bicycle Company Contribution Income Statement For the Month of June</th></tr><tr><th></th><th>Total</th><th>Per Unit</th></tr><tr><td>Sales (400 bicycles)</td><td>\$ 200,000</td><td>\$ 500</td></tr><tr><td>Less: Variables expenses</td><td>120,000</td><td>300</td></tr><tr><td>Contribution margin</td><td>80,000</td><td>\$ 200</td></tr><tr><td>Less: Fixed expenses</td><td>80,000</td><td></td></tr><tr><td>Net income</td><td>\$ -</td><td></td></tr></table>								Racing Bicycle Company Contribution Income Statement For the Month of June				Total	Per Unit	Sales (400 bicycles)	\$ 200,000	\$ 500	Less: Variables expenses	120,000	300	Contribution margin	80,000	\$ 200	Less: Fixed expenses	80,000		Net income	\$ -	
Racing Bicycle Company Contribution Income Statement For the Month of June																													
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The Contribution Approach

If Racing sells one more bike (401 bikes), net operating income will increase by \$200.



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Racing Bicycle Company Contribution Income Statement For the Month of June		
	Total	Per Unit
Sales (401 bicycles)	\$200,500	\$ 500
Less: Variables expenses	120,300	300
Contribution margin	80,200	\$ 200
Less: Fixed expenses	80,000	
Net income	\$ 200	

The Contribution Approach

We do not need to prepare an income statement to estimate profits at a particular sales volume. Simply multiply the number of units sold above break-even by the contribution margin per unit.



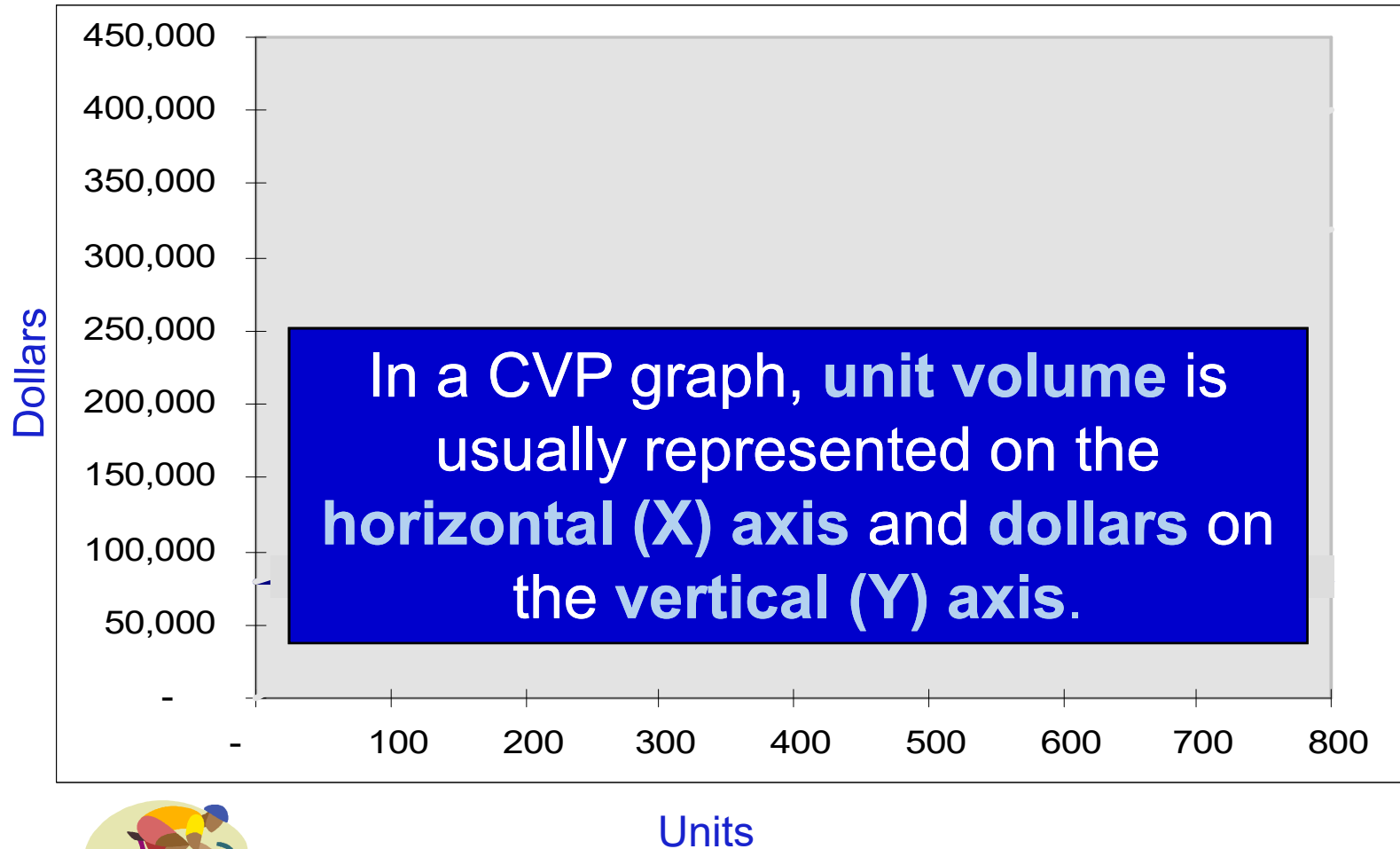
CVP Relationships in Graphic Form

The relationship among revenue, cost, profit and volume can be expressed graphically by preparing a CVP graph. Racing developed contribution margin income statements at 300, 400, and 500 units sold. We will use this information to prepare the CVP graph.

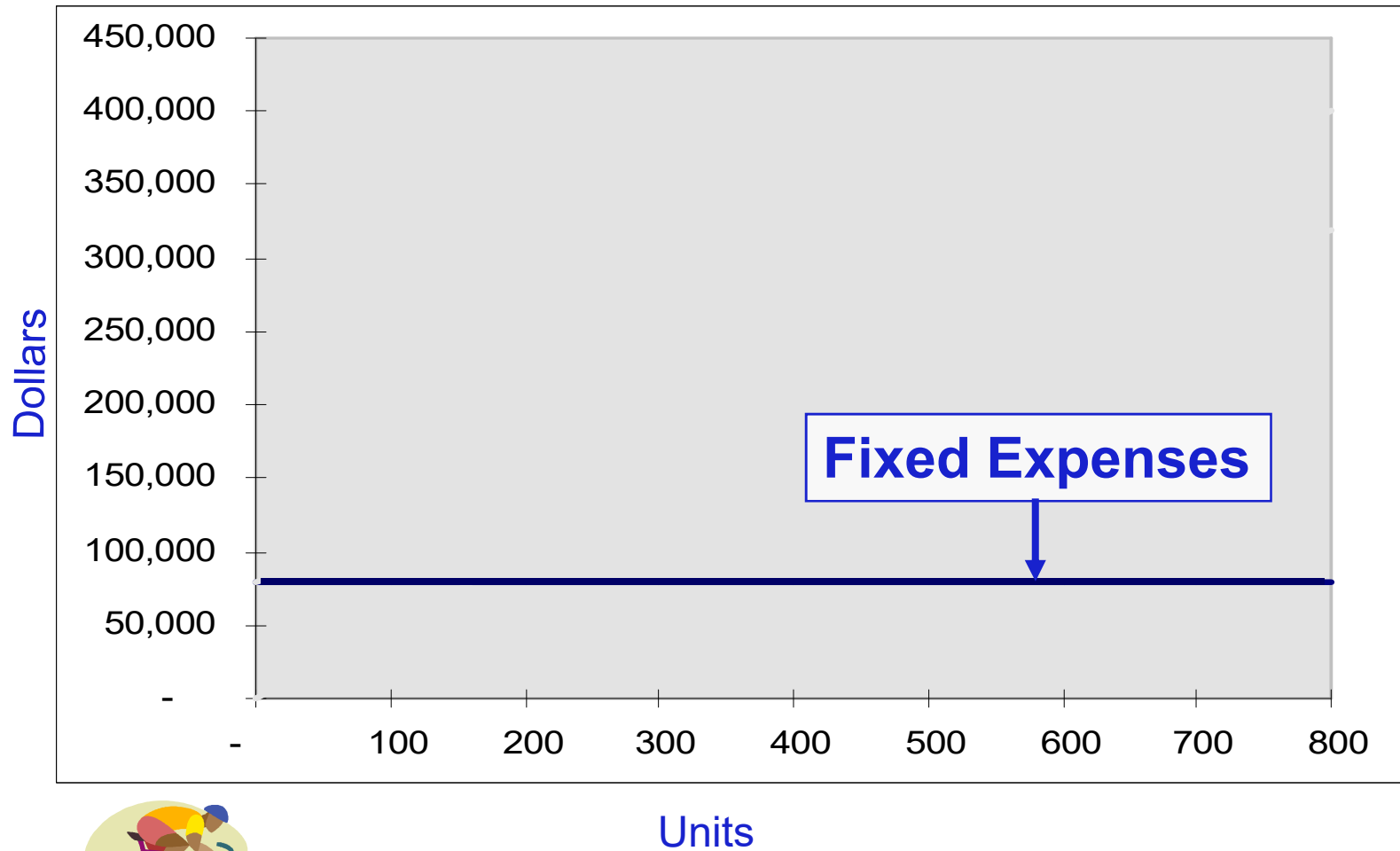
	Income 300 units	Income 400 units	Income 500 units
Sales	\$ 150,000	\$ 200,000	\$ 250,000
Less: variable expenses	90,000	120,000	150,000
Contribution margin	\$ 60,000	\$ 80,000	\$ 100,000
Less: fixed expenses	80,000	80,000	80,000
Net operating income	\$ (20,000)	\$ -	\$ 20,000



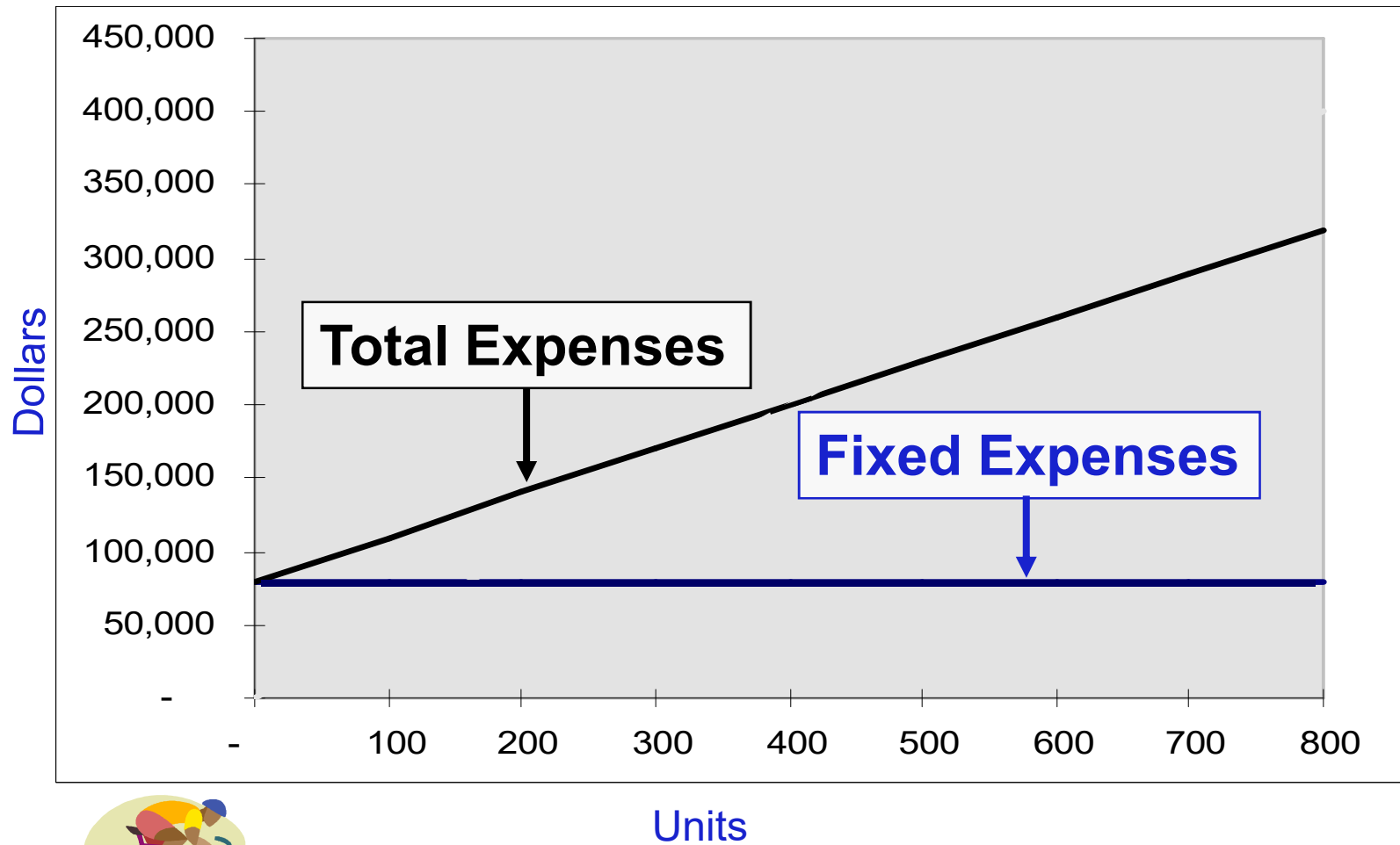
CVP Graph



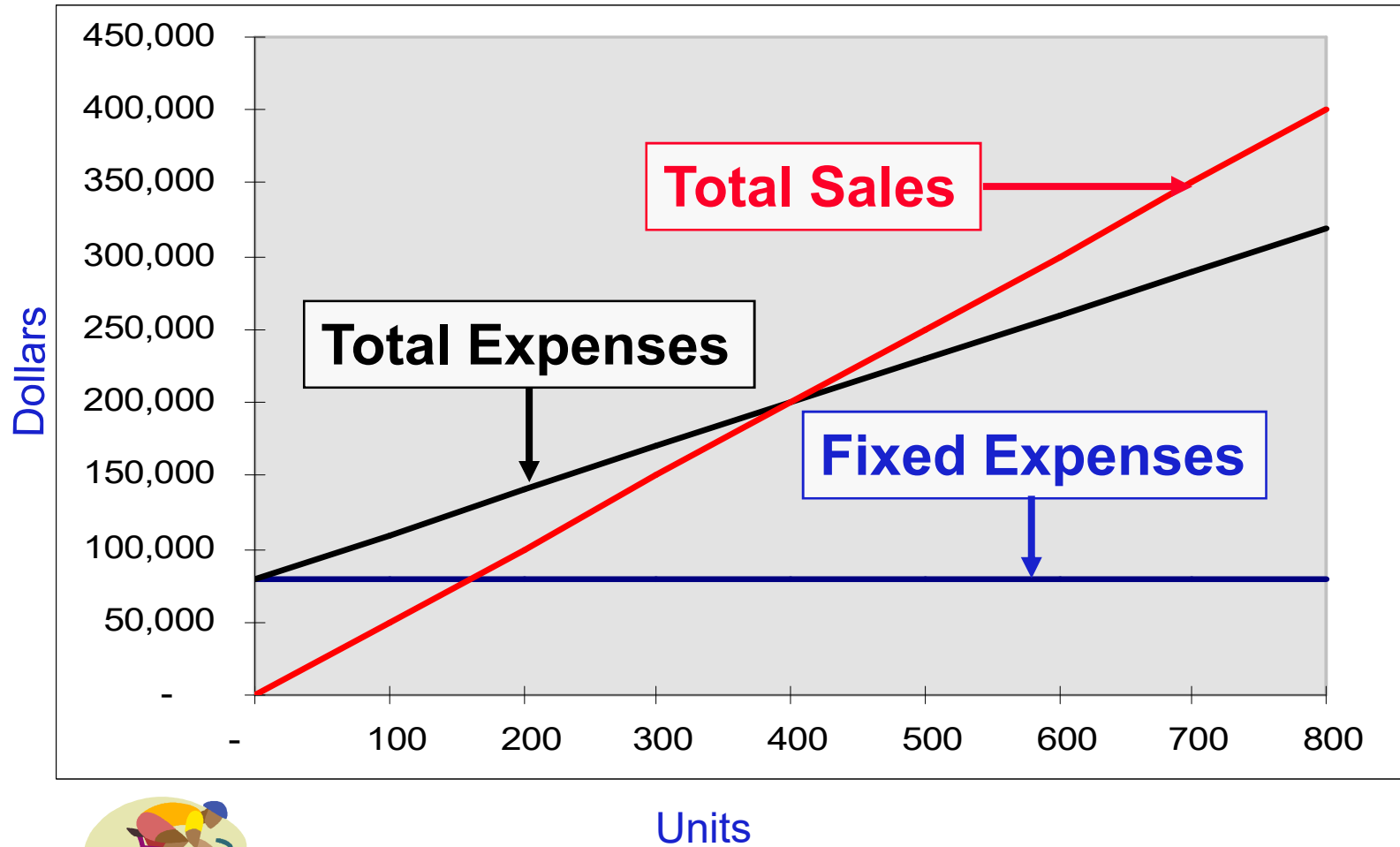
CVP Graph



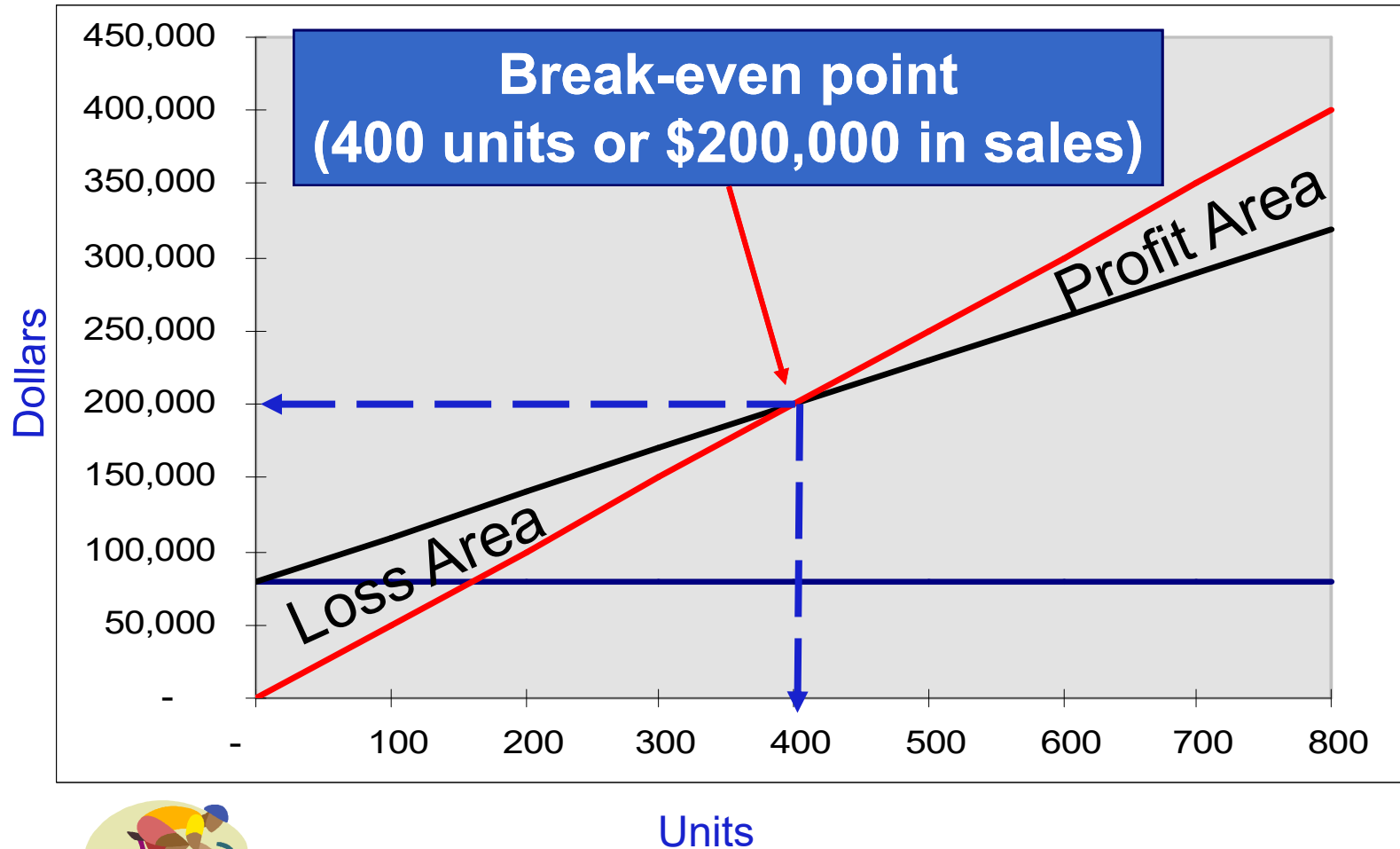
CVP Graph



CVP Graph



CVP Graph



Contribution Margin Ratio

The contribution margin **ratio** is:

$$\text{CM Ratio} = \frac{\text{Total CM}}{\text{Total sales}}$$

For Racing Bicycle Company the ratio is:

$$\frac{\$80,000}{\$200,000} = 40\%$$

Each \$1.00 increase in sales results in a total contribution margin increase of 40¢.



Contribution Margin Ratio

Or, in terms of **units**, the contribution margin **ratio** is:

$$\text{CM Ratio} = \frac{\text{Unit CM}}{\text{Unit selling price}}$$

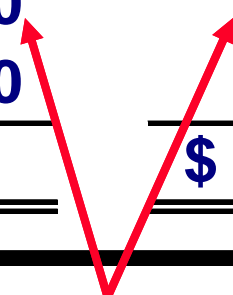
For Racing Bicycle Company the ratio is:

$$\frac{\$200}{\$500} = 40\%$$



Contribution Margin Ratio

	400 Bikes	500 Bikes
Sales	\$ 200,000	\$ 250,000
Less: variable expenses	120,000	150,000
Contribution margin	80,000	100,000
Less: fixed expenses	80,000	80,000
Net operating income	\$ -	\$ 20,000



A \$50,000 increase in sales revenue results in a \$20,000 increase in CM.
($\$50,000 \times 40\% = \$20,000$)

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. 2,100 cups are sold each month on average. What is the CM Ratio for Coffee Klatch?

- a. 1.319
- b. 0.758
- c. 0.242
- d. 4.139

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. 2,100 cups are sold each month on average. What is the CM Ratio for Coffee Klatch?

- a. 1.319
- ☒ b. 0.758
- c. 0.242
- d. 4.139

$$\begin{aligned}\text{CM Ratio} &= \frac{\text{Unit contribution margin}}{\text{Unit selling price}} \\ &= \frac{(\$1.49 - \$0.36)}{\$1.49} \\ &= \frac{\$1.13}{\$1.49} = 0.758\end{aligned}$$

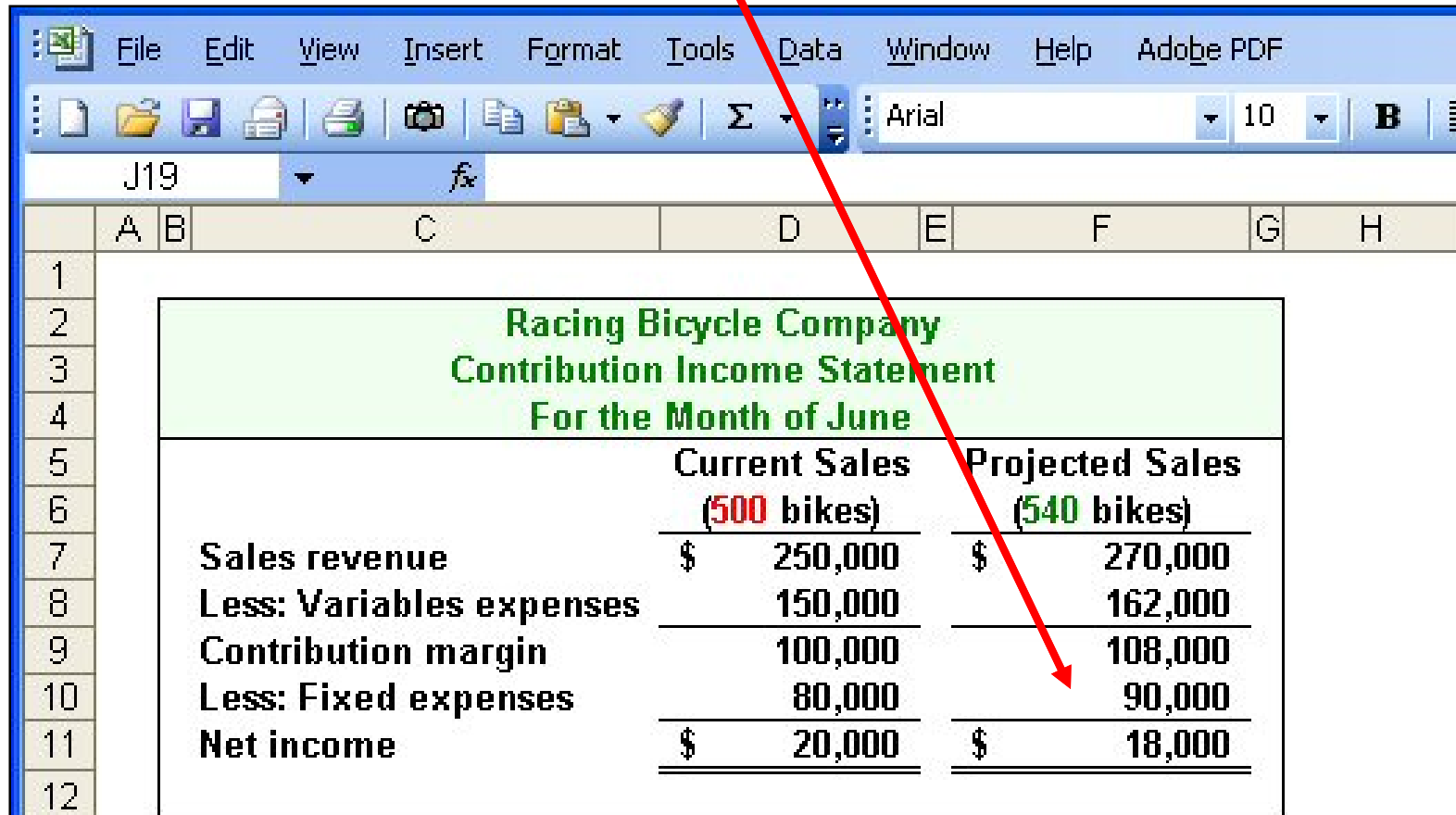
Changes in Fixed Costs and Sales Volume

What is the profit impact if Racing can increase unit sales from 500 to 540 by increasing the monthly advertising budget by \$10,000?



Changes in Fixed Costs and Sales Volume

\$80,000 + \$10,000 advertising = \$90,000



Racing Bicycle Company Contribution Income Statement For the Month of June							
	Current Sales (500 bikes)		Projected Sales (540 bikes)				
Sales revenue	\$	250,000	\$	270,000			
Less: Variables expenses		150,000		162,000			
Contribution margin		100,000		108,000			
Less: Fixed expenses		80,000		90,000			
Net income	\$	20,000	\$	18,000			

Sales *increased* by \$20,000, but net operating income *decreased* by \$2,000.

Changes in Fixed Costs and Sales Volume

The Shortcut Solution

Increase in CM (40 units X \$200)	\$ 8,000
Increase in advertising expenses	10,000
Decrease in net operating income	<u><u>\$ (2,000)</u></u>

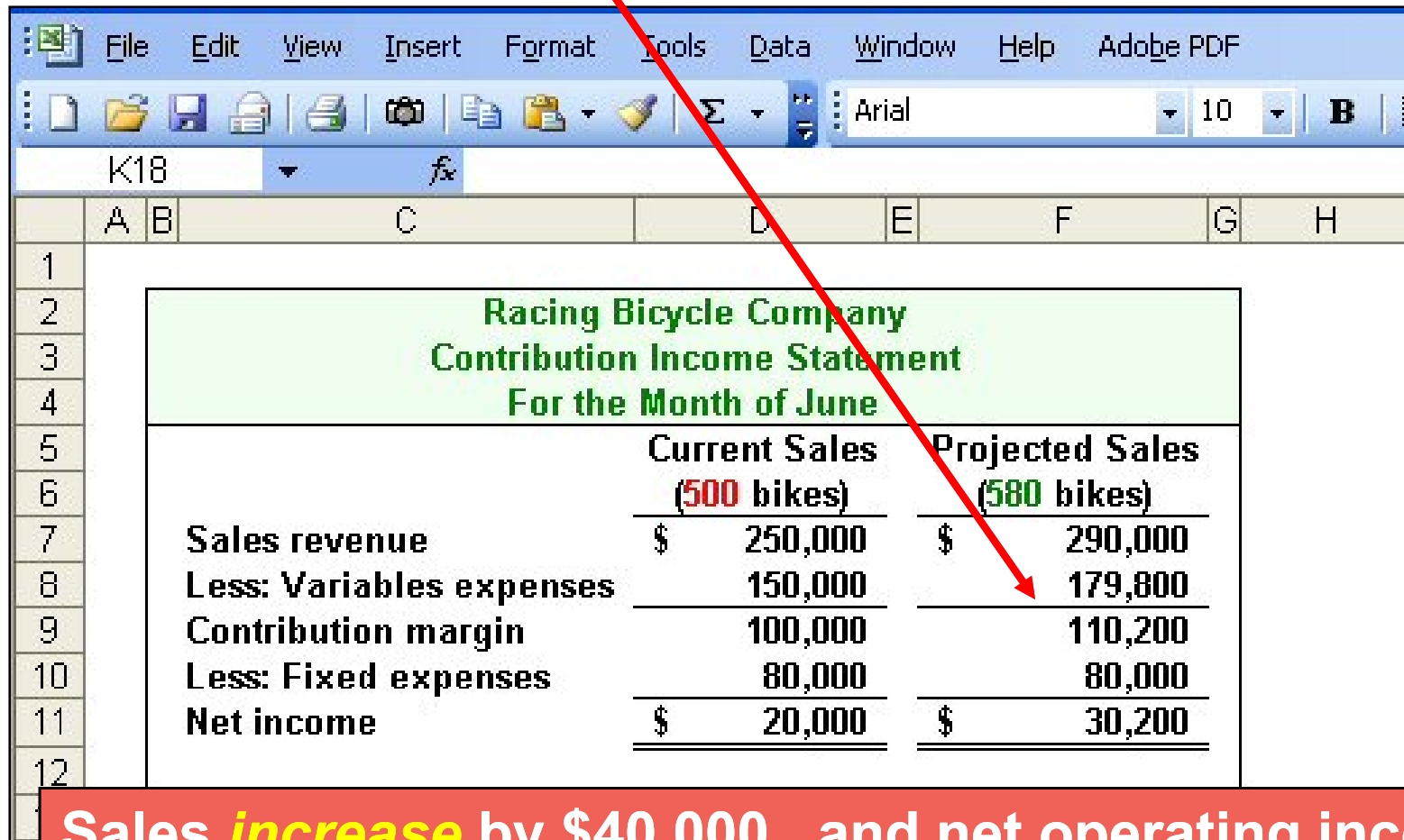
Change in Variable Costs and Sales Volume

What is the profit impact if Racing can use higher quality raw materials, thus increasing variable costs per unit by \$10, to generate an increase in unit sales from 500 to 580?



Change in Variable Costs and Sales Volume

$$580 \text{ units} \times \$310 \text{ variable cost/unit} = \$179,800$$



Racing Bicycle Company Contribution Income Statement For the Month of June							
	Current Sales (500 bikes)		Projected Sales (580 bikes)				
Sales revenue	\$	250,000	\$	290,000			
Less: Variables expenses		150,000		179,800			
Contribution margin		100,000		110,200			
Less: Fixed expenses		80,000		80,000			
Net income	\$	20,000	\$	30,200			

Sales **increase** by \$40,000, and net operating income **increases** by \$10,200.

Change in Fixed Cost, Sales Price and Volume

What is the profit impact if Racing (1) cuts its selling price \$20 per unit, (2) increases its advertising budget by \$15,000 per month, and (3) increases unit sales from 500 to 650 units per month?



Change in Fixed Cost, Sales Price and Volume

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1	<table><tr><th colspan="3">Racing Bicycle Company Contribution Income Statement For the Month of June</th></tr><tr><th></th><th>Current Sales (500 bikes)</th><th>Projected Sales (650 bikes)</th></tr><tr><td>Sales revenue</td><td>\$ 250,000</td><td>\$ 312,000</td></tr><tr><td>Less: Variables expenses</td><td>150,000</td><td>195,000</td></tr><tr><td>Contribution margin</td><td>100,000</td><td>117,000</td></tr><tr><td>Less: Fixed expenses</td><td>80,000</td><td>95,000</td></tr><tr><td>Net income</td><td>\$ 20,000</td><td>\$ 22,000</td></tr></table>								Racing Bicycle Company Contribution Income Statement For the Month of June				Current Sales (500 bikes)	Projected Sales (650 bikes)	Sales revenue	\$ 250,000	\$ 312,000	Less: Variables expenses	150,000	195,000	Contribution margin	100,000	117,000	Less: Fixed expenses	80,000	95,000	Net income	\$ 20,000	\$ 22,000
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Sales **increase** by \$62,000, fixed costs increase by \$15,000, and net operating income **increases** by \$2,000.

Change in Variable Cost, Fixed Cost and Sales Volume

What is the profit impact if Racing (1) pays a \$15 sales commission per bike sold instead of paying salespersons flat salaries that currently total \$6,000 per month, and (2) increases unit sales from 500 to 575 bikes?



Change in Variable Cost, Fixed Cost and Sales Volume

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Racing Bicycle Company Contribution Income Statement For the Month of June		
	Current Sales (500 bikes)	Projected Sales (575 bikes)
Sales revenue	\$ 250,000	\$ 287,500
Less: Variables expenses	150,000	181,125
Contribution margin	100,000	106,375
Less: Fixed expenses	80,000	74,000
Net income	\$ 20,000	\$ 32,375

Sales **increase** by \$37,500, variable costs **increase** by \$31,125, but fixed expenses **decrease** by \$6,000.

Change in Regular Sales Price

If Racing has an opportunity to sell 150 bikes to a wholesaler without disturbing sales to other customers or fixed expenses, what price would it quote to the wholesaler if it wants to increase monthly profits by \$3,000?



Change in Regular Sales Price

$$\begin{array}{rcl} \$ 3,000 \div 150 \text{ bikes} & = & \$ 20 \text{ per bike} \\ \text{Variable cost per bike} & = & 300 \text{ per bike} \\ \text{Selling price required} & = & \underline{\underline{\$ 320 \text{ per bike}}} \end{array}$$

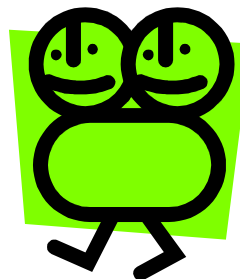
$$\begin{array}{rcl} 150 \text{ bikes} \times \$320 \text{ per bike} & = & \$ 48,000 \\ \text{Total variable costs} & = & 45,000 \\ \text{Increase in net income} & = & \underline{\underline{\$ 3,000}} \end{array}$$



Break-Even Analysis

Break-even analysis can be approached in two ways:

1. Equation method
2. Contribution margin method



Equation Method

Profits = (Sales – Variable expenses) – Fixed expenses

OR

Sales = Variable expenses + Fixed expenses + Profits

**At the break-even point
profits equal zero**



Break-Even Analysis

Here is the information from Racing Bicycle Company:

	<u>Total</u>	<u>Per Unit</u>	<u>Percent</u>
Sales (500 bikes)	\$ 250,000	\$ 500	100%
Less: variable expenses	150,000	300	60%
Contribution margin	\$ 100,000	\$ 200	40%
Less: fixed expenses	80,000		
Net operating income	\$ 20,000		



Equation Method

We calculate the break-even point as follows:

$$\text{Sales} = \text{Variable expenses} + \text{Fixed expenses} + \text{Profits}$$

$$\$500Q = \$300Q + \$80,000 + \$0$$

Where:

Q = Number of bikes sold

\$500 = Unit selling price

\$300 = Unit variable expense

\$80,000 = Total fixed expense



Equation Method

We calculate the break-even point as follows:

$$\text{Sales} = \text{Variable expenses} + \text{Fixed expenses} + \text{Profits}$$

$$\$500Q = \$300Q + \$80,000 + \$0$$

$$\$200Q = \$80,000$$

$$Q = \$80,000 \div \$200 \text{ per bike}$$

$$Q = 400 \text{ bikes}$$



Equation Method

The equation can be modified to calculate the break-even point in sales dollars.

$$\text{Sales} = \text{Variable expenses} + \text{Fixed expenses} + \text{Profits}$$

$$X = 0.60X + \$80,000 + \$0$$

Where:

X = Total sales dollars

0.60 = Variable expenses as a % of sales

\$80,000 = Total fixed expenses



Equation Method

The equation can be modified to calculate the break-even point in sales dollars.

$$\text{Sales} = \text{Variable expenses} + \text{Fixed expenses} + \text{Profits}$$

$$X = 0.60X + \$80,000 + \$0$$

$$0.40X = \$80,000$$

$$X = \$80,000 \div 0.40$$

$$X = \$200,000$$



Contribution Margin Method

The contribution margin method has two key equations.

$$\text{Break-even point in units sold} = \frac{\text{Fixed expenses}}{\text{Unit contribution margin}}$$

$$\text{Break-even point in total sales dollars} = \frac{\text{Fixed expenses}}{\text{CM ratio}}$$



Contribution Margin Method

Let's use the contribution margin method to calculate the break-even point in total sales dollars at Racing.

$$\text{Break-even point in total sales dollars} = \frac{\text{Fixed expenses}}{\text{CM ratio}}$$

$$\frac{\$80,000}{40\%} = \$200,000 \text{ break-even sales}$$



Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. 2,100 cups are sold each month on average. What is the break-even sales in units?

- a. 872 cups
- b. 3,611 cups
- c. 1,200 cups
- d. 1,150 cups

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49, and the average variable expense per cup is \$0.36, and the average fixed expense per month is \$1,300. How many cups must be sold each month on average to break even on sales in units?

- a. 872 cups
- b. 3,611 cups
- c. 1,200 cups
- d. 1,150 cups**

$$\begin{aligned}\text{Break-even} &= \frac{\text{Fixed expenses}}{\text{Unit CM}} \\ &= \frac{\$1,300}{\$1.49/\text{cup} - \$0.36/\text{cup}} \\ &= \frac{\$1,300}{\$1.13/\text{cup}} \\ &= \mathbf{1,150 \text{ cups}}\end{aligned}$$

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. 2,100 cups are sold each month on average. What is the break-even sales in dollars?

- a. \$1,300
- b. \$1,715
- c. \$1,788
- d. \$3,129

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. 2,100 cups are sold each month on average. What is the break-even sales in dollars?

a. \$1,300

b. \$1,715

c. \$1,788

d. \$3,129

$$\begin{aligned}\text{Break-even sales} &= \frac{\text{Fixed expenses}}{\text{CM Ratio}} \\ &= \frac{\$1,300}{0.758} \\ &= \mathbf{\$1,715}\end{aligned}$$

Target Profit Analysis

The equation and contribution margin methods can be used to determine the sales volume needed to achieve a target profit.

Suppose Racing Bicycle Company wants to know how many bikes must be sold to earn a profit of \$100,000.



The CVP Equation Method

$$\text{Sales} = \text{Variable expenses} + \text{Fixed expenses} + \text{Profits}$$

$$\$500Q = \$300Q + \$80,000 + \$100,000$$

$$\$200Q = \$180,000$$

$$Q = 900 \text{ bikes}$$



The Contribution Margin Approach

The contribution margin method can be used to determine that 900 bikes must be sold to earn the target profit of \$100,000.

$$\text{Unit sales to attain the target profit} = \frac{\text{Fixed expenses} + \text{Target profit}}{\text{Unit contribution margin}}$$

$$\frac{\$80,000 + \$100,000}{\$200/\text{bike}} = 900 \text{ bikes}$$



Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. How many cups of coffee would have to be sold to attain target profits of \$2,500 per month?

- a. 3,363 cups
- b. 2,212 cups
- c. 1,150 cups
- d. 4,200 cups

Quick Check ✓

$$\begin{aligned}\text{Unit sales to attain target profit} &= \frac{\text{Fixed expenses} + \text{Target profit}}{\text{Unit CM}} \\ &= \frac{\$1,300 + \$2,500}{\$1.49 - \$0.36} \\ &= \frac{\$3,800}{\$1.13} \\ &= 3,363 \text{ cups}\end{aligned}$$

- Contribution margin
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of \$2,5
- a. 3,363 cups
 - b. 2,212 cups
 - c. 1,150 cups
 - d. 4,200 cups

The Margin of Safety



The margin of safety is the excess of budgeted (or actual) sales over the break-even volume of sales.

Margin of safety = Total sales - Break-even sales

Let's look at Racing Bicycle Company and determine the margin of safety.



The Margin of Safety

If we assume that Racing Bicycle Company has actual sales of \$250,000, given that we have already determined the break-even sales to be \$200,000, the margin of safety is \$50,000 as shown

	Break-even sales 400 units	Actual sales 500 units
Sales	\$ 200,000	\$ 250,000
Less: variable expenses	120,000	150,000
Contribution margin	80,000	100,000
Less: fixed expenses	80,000	80,000
Net operating income	\$ -	\$ 20,000

The Margin of Safety

The margin of safety can be expressed as
20% of sales.

$$(\$50,000 \div \$250,000)$$

	Break-even sales 400 units	Actual sales 500 units
Sales	\$ 200,000	\$ 250,000
Less: variable expenses	120,000	150,000
Contribution margin	80,000	100,000
Less: fixed expenses	80,000	80,000
Net operating income	\$ -	\$ 20,000

The Margin of Safety

The margin of safety can be expressed in terms of the number of units sold. The margin of safety at Racing is \$50,000, and each bike sells for \$500.

$$\text{Margin of Safety in units} = \frac{\$50,000}{\$500} = 100 \text{ bikes}$$



Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. 2,100 cups are sold each month on average. What is the margin of safety?

- a. 3,250 cups
- b. 950 cups
- c. 1,150 cups
- d. 2,100 cups

Quick Check ✓

Coffee Klatch is an espresso stand in a

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$$\begin{aligned}\text{Margin of safety} &= \text{Total sales} - \text{Break-even sales} \\ &= 2,100 \text{ cups} - 1,150 \text{ cups} \\ &= \mathbf{950 \text{ cups}}\end{aligned}$$

or

$$\begin{aligned}\text{Margin of safety} &= \frac{950 \text{ cups}}{2,100 \text{ cups}} = \mathbf{45\%} \\ \text{percentage}\end{aligned}$$

a. \$

b. 950 cups

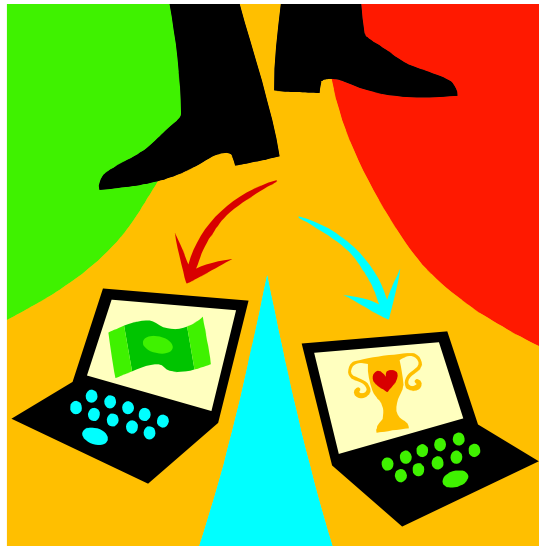
c. 1,150 cups

d. 2,100 cups

Cost Structure and Profit Stability

Cost structure refers to the relative proportion of fixed and variable costs in an organization.

Managers often have some latitude in determining their organization's cost structure.



Cost Structure and Profit Stability

There are advantages and disadvantages to high fixed cost (or low variable cost) and low fixed cost (or high variable cost) structures.

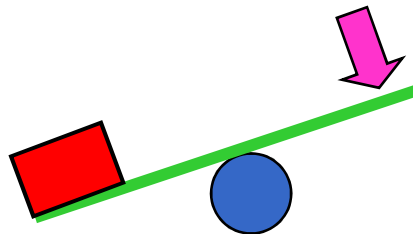
An advantage of a high fixed cost structure is that income will be higher in good years compared to companies with lower proportion of fixed costs.

A disadvantage of a high fixed cost structure is that income will be lower in bad years compared to companies with lower proportion of fixed costs.

Operating Leverage

- A measure of how sensitive net operating income is to percentage changes in sales.

$$\text{Degree of operating leverage} = \frac{\text{Contribution margin}}{\text{Net operating income}}$$



Operating Leverage

At Racing, the degree of operating leverage is 5.

	Actual sales
	500 Bikes
	\$ 250,000
	Less: variable expenses
Sales	150,000
Less: variable expenses	
Contribution margin	100,000
Less: fixed expenses	
Net income	80,000
	\$ 20,000

$$\frac{\$100,000}{\$20,000} = 5$$

Operating Leverage

With an operating leverage of 5, if Racing increases its sales by 10%, net operating income would increase by 50%.

Percent increase in sales		10%
Degree of operating leverage	×	<u>5</u>
Percent increase in profits		<u><u>50%</u></u>

Here's the verification!



Operating Leverage

	Actual sales (500)	Increased sales (550)
Sales	\$ 250,000	\$ 275,000
Less variable expenses	150,000	165,000
Contribution margin	100,000	110,000
Less fixed expenses	80,000	80,000
Net operating income	\$ 20,000	\$ 30,000

**10% increase in sales from
\$250,000 to \$275,000 . . .**

**. . . results in a 50% increase in
income from \$20,000 to \$30,000.**

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. 2,100 cups are sold each month on average. What is the operating leverage?

- a. 2.21
- b. 0.45
- c. 0.34
- d. 2.92

Quick Check ✓

Coffee Klatch is a downtown office building selling price of a cup of coffee is \$0.36. The average variable cost is \$0.36. The average fixed cost is \$0.36. The average contribution margin is \$1,300. 2,100 cups are sold on average. What is the operating leverage?

- ☒ a. 2.21
- ☐ b. 0.45
- ☐ c. 0.34
- ☐ d. 2.92

	<i>Actual sales 2,100 cups</i>
Sales	\$ 3,129
Less: Variable expenses	756
Contribution margin	2,373
Less: Fixed expenses	1,300
Net operating income	\$ 1,073

$$\begin{aligned}\text{Operating leverage} &= \frac{\text{Contribution margin}}{\text{Net operating income}} \\ &= \frac{\$2,373}{\$1,073} = 2.21\end{aligned}$$

Quick Check ✓

At Coffee Klatch the average selling price of a cup of coffee is \$1.49, the average variable expense per cup is \$0.36, and the average fixed expense per month is \$1,300. 2,100 cups are sold each month on average.

If sales increase by 20%, by how much should net operating income increase?

- a. 30.0%
- b. 20.0%
- c. 22.1%
- d. 44.2%

Quick Check ✓

At Coffee Klatch the average selling price of a cup of coffee is \$1.49, the average variable expense per cup is \$0.36, and the average fixed expense per month is \$1,300. 2,100 cups are sold each month on average.

If sales increase by 20%, by how much should net operating income increase?

a. 30.0%

b. 20.0%

c. 22.1%

d. 44.2%

Percent increase in sales	20.0%
× Degree of operating leverage	2.21
Percent increase in profit	44.20%

Verify Increase in Profit

	<i>Actual sales</i>	<i>Increased sales</i>
	<i>2,100 cups</i>	<i>2,520 cups</i>
Sales	\$ 3,129	\$ 3,755
Less: Variable expenses	756	907
Contribution margin	2,373	2,848
Less: Fixed expenses	1,300	1,300
Net operating income	\$ 1,073	\$ 1,548
% change in sales		20.0%
% change in net operating income		44.2%

Structuring Sales Commissions

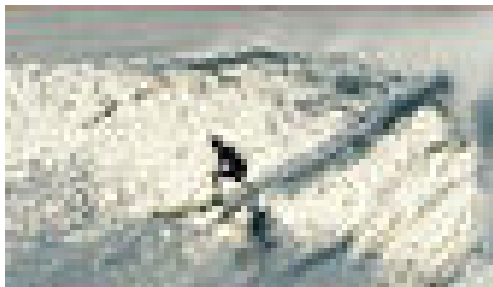
Companies generally compensate salespeople by paying them either a commission based on sales or a salary plus a sales commission. Commissions based on sales dollars can lead to lower profits in a company.

Let's look at an example.

Structuring Sales Commissions

Pipeline Unlimited produces two types of surfboards, the XR7 and the Turbo. The XR7 sells for \$100 and generates a contribution margin per unit of \$25. The Turbo sells for \$150 and earns a contribution margin per unit of \$18.

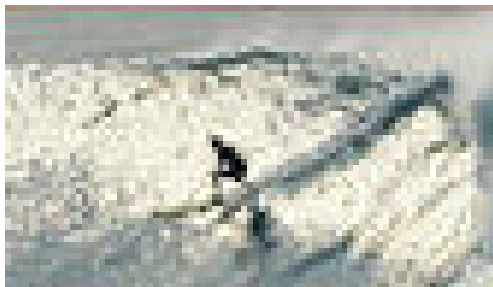
The sales force at Pipeline Unlimited is compensated based on sales commissions.



Structuring Sales Commissions

If you were on the sales force at Pipeline, you would push hard to sell the Turbo even though the XR7 earns a higher contribution margin per unit.

To eliminate this type of conflict, commissions can be based on contribution margin rather than on selling price alone.



The Concept of Sales Mix

- Sales mix is the relative proportion in which a company's products are sold.
- Different products have different selling prices, cost structures, and contribution margins.

Let's assume Racing Bicycle Company sells bikes and carts and that the sales mix between the two products remains the same.



Multi-product break-even analysis

Racing Bicycle Co. provides the following information:

S23 fx															
	AB	C	D	E	F	G	H	I	J	K	L	M	N	O	
1															
2															
3															
4															
5															
6															
7															
8															
9															
10															

	Bicycles			Carts			Total	
Sales	\$ 250,000	100%		\$ 300,000	100%		\$ 550,000	100.0%
Variable expenses	150,000	60%		135,000	45%		285,000	51.8%
Contribution margin	<u>\$ 100,000</u>	<u>40%</u>		<u>\$ 165,000</u>	<u>55%</u>		<u>265,000</u>	<u>48.2%</u>
Fixed expenses							170,000	
Net operating income							<u>\$ 95,000</u>	
Sales mix	\$ 250,000	45%		\$ 300,000	55%		\$ 550,000	100%

$$\frac{\$265,000}{\$550,000} = 48.2\% \text{ (rounded)}$$

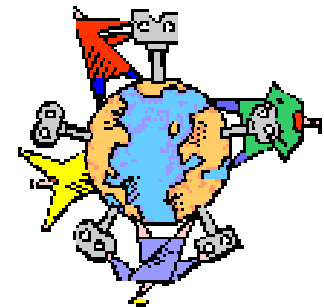
Multi-product break-even analysis

$$\begin{aligned}
 \text{Break-even sales} &= \frac{\text{Fixed expenses}}{\text{CM Ratio}} \\
 &= \frac{\$170,000}{48.2\%} \\
 &= \$352,697
 \end{aligned}$$

File Edit View Insert Format Tools Data Window Help Adobe PDF														
S25 fx Arial 10 B														
	AB	C	D	E	F	G	H	I	J	K	L	M	N	O
1														
2														
3	Sales		\$ 158,714	100%	\$ 193,983	100%	\$ 352,697	100.0%						
4	Variable expenses		95,228	60%	87,292	45%	182,521	51.8%						
5	Contribution margin		\$ 63,486	40%	\$ 106,691	55%	170,176	48.2%						
6	Fixed expenses							170,000						
7	Net operating income							Rounding error → \$ 176						
8														
9	Sales mix		\$ 158,714	45%	\$ 193,983	55%	\$ 352,697	100%						

Key Assumptions of CVP Analysis

- 1 Selling price is constant.
- 2 Costs are linear.
- 3 In multi-product companies, the sales mix is constant.
- 4 In manufacturing companies, inventories do not change (units produced = units sold).



End of Chapter 6

